

Rank Preservation, Exact-Recovery Obstructions, and Exceptional Strict-Saddle Trajectories in Robust Matrix Sensing

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Status. This paper gives *partial progress, not a complete solution* of the active/non-active strict-saddle question posed by Ma and Fattahi. It proves a negative exact-convergence result for full-dimensional initializations under summable nonsingular updates, and a positive existence result for exceptional rank- r trajectories.

Abstract

We revisit a proposed invariant-face argument for the nonsmooth ℓ_1 formulation of symmetric robust matrix sensing. The invariant-face observation is correct, but it does not establish positive-probability attraction from a full-dimensional random initialization. We prove a stronger structural theorem. Every Clarke subgradient of the factorized loss has the form $G(W)W$, so each subgradient update is a left multiplication. For any trajectory

$$W_{t+1} = (I - H_t)W_t$$

with $\sum_t \|H_t\|_{\text{op}} < \infty$, the multiplier product converges. If no multiplier $I - H_t$ is singular, the limiting multiplier is invertible; hence the factor rank and right kernel are preserved even at the limit. A quantitative version yields uniform positive lower bounds on both the factor distance and the matrix-recovery error whenever the initialization has rank larger than the ground-truth rank. The result applies directly to ordinary geometrically decaying SubGM and to the normalized exponentially decaying rule analyzed by Ma and Fattahi, under an explicit small-initial-step condition.

Consequently, in exact arithmetic, an absolutely continuous full-dimensional initialization cannot converge exactly to the symmetric rank- r true-factor set in this regime. Conversely, every rank- r right-orthogonal face is invariant. We verify that published Gaussian landscape and exact-rank local-convergence theorems can be combined to produce exceptional deterministic trajectories converging to strict-saddle true factors, and we give an explicit 2×2 robust sensing instance with one corrupted measurement for which an arbitrarily small exact-rank initialization converges to a strict saddle. These trajectories lie in a measure-zero determinantal variety. Thus the original argument can be repaired only as partial progress: it does not settle active versus non-active saddle geometry or positive-measure attraction.

1 Introduction

Let $X_\star \in \mathbb{R}^{d \times d}$ be positive semidefinite with $\text{rank}(X_\star) = r$, and let $k > r$ be the search rank. The symmetric robust matrix-sensing objective is

$$F_k(W) := \frac{1}{m} \sum_{i=1}^m \left| \left\langle A_i, WW^\top \right\rangle - y_i \right|, \quad W \in \mathbb{R}^{d \times k}. \quad (1)$$

Its true-factor set is

$$\mathcal{W}_k := \{W \in \mathbb{R}^{d \times k} : WW^\top = X_\star\}. \quad (2)$$

Ma and Fattahi proved that, in an overparameterized noisy Gaussian regime, true factors can be Clarke-critical strict saddles rather than local minima [1]. They also reported experiments in which SubGM, initialized by a full-dimensional Gaussian perturbation of a true factor and run with an exponentially decaying step, appeared to converge to the true-factor set. They proposed two possible explanations: true factors may be non-active strict saddles, or random initialization may be insufficient for SubGM to escape even active strict saddles. They left proof or refutation open.

A proposed resolution embedded the exact-rank problem into the face

$$\{[U, 0] : U \in \mathbb{R}^{d \times r}\} \subset \mathbb{R}^{d \times k}$$

and observed that this face is invariant. It then imported an exact-rank convergence theorem and concluded that SubGM can converge to strict saddles with positive probability. Three distinct issues are hidden in that conclusion:

- (i) invariance of a lower-dimensional face is not transverse attraction;
- (ii) probability over sensing instances is not probability over a full-dimensional initialization;
- (iii) the cited “small initialization” theorem of [2] is an accuracy-parametrized finite-time guarantee, whereas exact asymptotic convergence from a fixed full-rank initialization is a different assertion.

This paper separates the valid and invalid parts of that argument. The main new observation is that the invariant-face lemma is a special case of a global multiplicative structure. Once combined with elementary facts about infinite matrix products, this gives an exact-recovery obstruction for summable steps.

Main conclusions.

- (a) Under summable effective increments and nonsingular update matrices, SubGM preserves the factor rank and right kernel at the limit.
- (b) A higher-rank initialization has a strictly positive error floor; hence a full-dimensional absolutely continuous initialization has zero probability of exact convergence to $\mathcal{W}_k$.
- (c) Rank- r faces nevertheless carry exceptional deterministic trajectories to strict saddles. We prove this both through a compatible use of existing Gaussian theorems and through an explicit two-dimensional example.
- (d) These results do not classify the true factors as active or non-active strict saddles and do not rule out finite-accuracy or positive-measure behavior for other step rules.

2 Clarke subgradients are multiplicative

For $a \in \mathbb{R}$, set

$$\text{Sign}(a) := \begin{cases} \{1\}, & a > 0, \\ [-1, 1], & a = 0, \\ \{-1\}, & a < 0. \end{cases}$$

Define

$$\rho_i(W) := \langle A_i, WW^\top \rangle - y_i, \quad B_i := A_i + A_i^\top,$$

and

$$L_A := \frac{1}{m} \sum_{i=1}^m \|B_i\|_{\text{op}}. \quad (3)$$

All matrices A_i and all data y_i are fixed throughout the deterministic results below.

Definition 2.1 (Strict saddle used here). *A point $\bar{W}$ is a strict saddle if $0 \in \partial_C F_k(\bar{W})$ and there are a unit-Frobenius direction V , constants $c > 0$ and $\gamma_0 > 0$, such that*

$$F_k(\bar{W} + \gamma V) - F_k(\bar{W}) \leq -c\gamma^2 \quad (0 < \gamma \leq \gamma_0).$$

This is the quadratic-descent notion used in the motivating landscape paper. It should not be confused with the stronger notion of an active strict saddle.

Proposition 2.2 (Exact Clarke subdifferential). *For every $W \in \mathbb{R}^{d \times k}$,*

$$\partial_C F_k(W) = \left\{ \frac{1}{m} \sum_{i=1}^m s_i B_i W : s_i \in \text{Sign}(\rho_i(W)) \right\}. \quad (4)$$

Consequently every $D \in \partial_C F_k(W)$ has the form

$$D = GW, \quad G = \frac{1}{m} \sum_{i=1}^m s_i B_i = G^\top, \quad \|G\|_{\text{op}} \leq L_A. \quad (5)$$

Proof. Let $\Sigma = \{-1, 1\}^m$. Since $\sum_i |a_i| = \max_{\sigma \in \Sigma} \sum_i \sigma_i a_i$,

$$F_k(W) = \max_{\sigma \in \Sigma} \phi_\sigma(W), \quad \phi_\sigma(W) := \frac{1}{m} \sum_{i=1}^m \sigma_i \rho_i(W).$$

Each ϕ_σ is C^1 and

$$\nabla \phi_\sigma(W) = \frac{1}{m} \sum_{i=1}^m \sigma_i B_i W.$$

For a finite maximum of C^1 functions, the Clarke subdifferential is the convex hull of the active gradients. At W , the sign σ_i is forced to equal $\text{sign}(\rho_i(W))$ when $\rho_i(W) \neq 0$, and both signs are active when $\rho_i(W) = 0$. The convex hull of the resulting hypercube of active sign vectors is exactly the set obtained by allowing $s_i \in [-1, 1]$ independently at zero residuals. This proves (4). Symmetry of G and the bound in (5) follow from the triangle inequality. $\square$

Consider a SubGM trajectory

$$W_{t+1} = W_t - \alpha_t D_t, \quad D_t \in \partial_C F_k(W_t), \quad \alpha_t \geq 0. \quad (6)$$

For a chosen representation $D_t = G_t W_t$, define

$$H_t := \alpha_t G_t, \quad M_t := I - H_t. \quad (7)$$

Then

$$W_{t+1} = M_t W_t. \quad (8)$$

The effective increment H_t , rather than the scalar α_t alone, is the relevant quantity.

Proposition 2.3 (Finite-time row-space monotonicity). *Every update of the form (8) satisfies*

$$\ker(W_t) \subseteq \ker(W_{t+1}), \quad \text{row}(W_{t+1}) \subseteq \text{row}(W_t), \quad \text{rank}(W_{t+1}) \leq \text{rank}(W_t).$$

If M_t is invertible, all three relations are equalities.

Proof. If $W_t z = 0$, then $W_{t+1} z = M_t W_t z = 0$. The row-space statement is the orthogonal-complement form of the kernel statement. If M_t is invertible, then $W_t = M_t^{-1} W_{t+1}$ gives the reverse inclusions. $\square$

3 A matrix-product theorem for summable updates

The next theorem is purely linear algebra and permits adaptive, nonsymmetric increments H_t .

Theorem 3.1 (Summable multiplier products). *Let $Z_0 \in \mathbb{R}^{d \times k}$ and*

$$Z_{t+1} = (I - H_t)Z_t, \quad a_t := \|H_t\|_{\text{op}}. \quad (9)$$

Assume

$$A_H := \sum_{t=0}^{\infty} a_t < \infty. \quad (10)$$

Set $P_0 = I$ and

$$P_t := (I - H_{t-1}) \cdots (I - H_0) \quad (t \geq 1).$$

Then:

(i) P_t converges in operator norm to a matrix P_∞ , and $Z_t \rightarrow Z_\infty := P_\infty Z_0$.

(ii) If every $I - H_t$ is nonsingular, then P_∞ is nonsingular. In that case

$$\ker(Z_t) = \ker(Z_0), \quad \text{rank}(Z_t) = \text{rank}(Z_0) \quad \text{for all } t \in \mathbb{N} \cup \{\infty\}.$$

(iii) If, more quantitatively,

$$q := \sup_{t \geq 0} a_t < 1, \quad (11)$$

then for every $t \in \mathbb{N} \cup \{\infty\}$,

$$\|P_t\|_{\text{op}} \leq e^{A_H}, \quad \sigma_{\min}(P_t) \geq c_H := \exp\left(-\frac{A_H}{1-q}\right) > 0. \quad (12)$$

Proof. For $n > m$ write

$$P_n = R_{n,m} P_m, \quad R_{n,m} := (I - H_{n-1}) \cdots (I - H_m).$$

Repeatedly expanding the product, or simply using submultiplicativity, gives

$$\|R_{n,m} - I\|_{\text{op}} \leq \prod_{t=m}^{n-1} (1 + a_t) - 1 \leq \exp\left(\sum_{t=m}^{n-1} a_t\right) - 1.$$

Also $\|P_m\|_{\text{op}} \leq \prod_{t < m} (1 + a_t) \leq e^{A_H}$. Hence

$$\|P_n - P_m\|_{\text{op}} \leq e^{A_H} \left[\exp\left(\sum_{t=m}^{n-1} a_t\right) - 1 \right] \rightarrow 0.$$

Thus $P_t \rightarrow P_\infty$ and $Z_t = P_t Z_0 \rightarrow P_\infty Z_0$.

For part (ii), $a_t \rightarrow 0$, so choose T such that $a_t \leq 1/2$ for $t \geq T$. Then

$$\sigma_{\min}(I - H_t) \geq 1 - a_t$$

and, for $n > T$,

$$\sigma_{\min}((I - H_{n-1}) \cdots (I - H_T)) \geq \prod_{t=T}^{n-1} (1 - a_t).$$

The infinite product on the right is positive because $\sum_{t \geq T} a_t < \infty$ and $a_t \leq 1/2$. The limiting tail product is therefore nonsingular. Multiplying by the nonsingular finite prefix proves that P_∞ is nonsingular. Kernel and rank preservation follow from $Z_t = P_t Z_0$.

For part (iii), if $0 \leq x \leq q < 1$, then

$$-\log(1 - x) = \int_0^x \frac{du}{1 - u} \leq \frac{x}{1 - q}.$$

Therefore

$$\sigma_{\min}(P_t) \geq \prod_{j < t} (1 - a_j) \geq \exp\left(-\frac{1}{1 - q} \sum_{j < t} a_j\right) \geq c_H.$$

The upper bound follows from $\prod_{j < t} (1 + a_j) \leq e^{A_H}$, and both bounds pass to the limit. $\square$

Remark 3.2 (Why summability matters). *The theorem does not say that rank can never fall. A singular multiplier can reduce rank in one step, and a nonsummable sequence can have an invertible product at every finite time but a singular limit. The conclusions are specific to summable effective increments together with nonsingularity.*

4 Consequences for symmetric robust matrix sensing

4.1 Ordinary geometric steps

For (6), $\|H_t\|_{\text{op}} \leq L_A \alpha_t$ by Theorem 2.2. Hence the abstract theorem immediately gives the following.

Corollary 4.1 (Ordinary geometrically decaying SubGM). *Let $\alpha_t = \eta \rho^t$ with $0 < \rho < 1$. If*

$$\eta L_A < 1, \tag{13}$$

then every trajectory of (6) converges, and its rank and right kernel agree with those of W_0 at every finite time and at the limit. Moreover, Theorem 3.1 applies with

$$A_H \leq \frac{\eta L_A}{1 - \rho}, \quad q \leq \eta L_A, \quad c_H \geq \exp\left[-\frac{\eta L_A}{(1 - \rho)(1 - \eta L_A)}\right]. \tag{14}$$

4.2 The normalized exponential rule of Ma–Fattahi

Write

$$Q_t := \frac{1}{m} \sum_{i=1}^m s_{i,t} A_i, \quad s_{i,t} \in \text{Sign}(\rho_i(W_t)). \tag{15}$$

Then a subgradient is $(Q_t + Q_t^\top)W_t$. The normalized exponential rule analyzed in [2] is, in their norm convention,

$$W_{t+1} = W_t - \frac{\eta\rho^t}{\|Q_t\|_{\text{op}}}(Q_t + Q_t^\top)W_t, \quad 0 < \rho < 1, \quad (16)$$

when $Q_t \neq 0$. If $Q_t = 0$, we use the natural convention that the update is zero. Notice that the scalar step in (16) need not be summable; nevertheless the *effective* matrix increment is.

Corollary 4.2 (Rank preservation for the normalized rule). *Suppose (16) is used and*

$$0 < \eta < \frac{1}{2}. \quad (17)$$

Then every trajectory converges, and rank and right kernel are preserved at the limit. Uniformly over every sensing instance and every admissible sign selection, one may take

$$A_H \leq \frac{2\eta}{1-\rho}, \quad q \leq 2\eta, \quad c_H \geq \exp\left[-\frac{2\eta}{(1-\rho)(1-2\eta)}\right]. \quad (18)$$

Proof. For $Q_t \neq 0$, the effective increment is

$$H_t = \eta\rho^t \frac{Q_t + Q_t^\top}{\|Q_t\|_{\text{op}}},$$

so

$$\|H_t\|_{\text{op}} \leq \eta\rho^t \frac{\|Q_t\|_{\text{op}} + \|Q_t^\top\|_{\text{op}}}{\|Q_t\|_{\text{op}}} \leq 2\eta\rho^t.$$

For $Q_t = 0$, $H_t = 0$. Apply Theorem 3.1. □

4.3 Uniform exact-recovery obstructions

For $W \in \mathbb{R}^{d \times k}$ define

$$\text{dist}_F(W, \mathcal{W}_k) := \inf_{Y \in \mathcal{W}_k} \|W - Y\|_F.$$

Theorem 4.3 (Factor and matrix error floors). *Suppose the trajectory has the form (9) and satisfies (10)–(11). Let $s := \text{rank}(W_0) > r$. Then, for every $t \in \mathbb{N} \cup \{\infty\}$,*

$$\text{dist}_F(W_t, \mathcal{W}_k) \geq c_H \left(\sum_{j=r+1}^s \sigma_j(W_0)^2 \right)^{1/2} > 0, \quad (19)$$

$$\left\| W_t W_t^\top - X_\star \right\|_F \geq c_H^2 \left(\sum_{j=r+1}^s \sigma_j(W_0)^4 \right)^{1/2} > 0. \quad (20)$$

If instead $\text{rank}(W_0) < r$ and all multipliers are nonsingular, exact convergence to $\mathcal{W}_k$ is likewise impossible.

Proof. By Theorem 3.1,

$$\sigma_j(W_t) = \sigma_j(P_t W_0) \geq \sigma_{\min}(P_t) \sigma_j(W_0) \geq c_H \sigma_j(W_0).$$

Every matrix in $\mathcal{W}_k$ has rank r . The Eckart–Young theorem and the inclusion $\mathcal{W}_k \subseteq \{Y : \text{rank}(Y) \leq r\}$ give

$$\text{dist}_F(W_t, \mathcal{W}_k) \geq \left(\sum_{j=r+1}^s \sigma_j(W_t)^2 \right)^{1/2},$$

which proves (19). The nonzero eigenvalues of $W_t W_t^\top$ are $\sigma_j(W_t)^2$. Since $X_\star$ has rank r , applying Eckart–Young to $W_t W_t^\top$ gives (20). Finally, when $\text{rank}(W_0) < r$, rank preservation rules out a rank- r limit. $\square$

Corollary 4.4 (Zero probability of exact convergence). *Assume $r < \min\{d, k\}$. Fix any sensing instance and any measurable subgradient-selection rule. Under either Theorem 4.1 or Theorem 4.2, let W_0 have a law absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^{d \times k}$. Then*

$$\mathbb{P}(\text{dist}_F(W_t, \mathcal{W}_k) \rightarrow 0) = 0, \quad \mathbb{P}\left(\left\|W_t W_t^\top - X_\star\right\|_F \rightarrow 0\right) = 0. \quad (21)$$

The assertion remains true after averaging over random sensing instances.

Proof. An absolutely continuous $d \times k$ random matrix has rank $\min\{d, k\} > r$ almost surely. Apply Theorem 4.3. Equivalently, the product theorem implies that W_t has a limit of the same rank as W_0 , while every true factor has rank r . $\square$

Remark 4.5 (Why a numerical trajectory can look convergent). *If $W_0 = W_\star + E$ with a tiny full-dimensional perturbation E , the singular values beyond rank r may be extremely small. The factor floor (19) is linear and the Gram floor (20) is quadratic in those trailing singular values, while c_H can itself be conservative. Both floors can therefore lie below a plotting tolerance or floating-point resolution. This is compatible with the finite-accuracy theorem of [2], which chooses the initialization scale as a function of a prescribed target accuracy and guarantees an accurate finite iterate. It is not an exact asymptotic recovery theorem for one fixed full-rank perturbation.*

5 Exceptional exact-rank trajectories

The negative result above does not prevent convergence from an initialization that already has rank r . We now state the invariant-face mechanism in its natural right-orthogonally invariant form.

Proposition 5.1 (Rotated invariant faces). *Let $1 \leq s \leq k$ and let $R \in \mathbb{R}^{s \times k}$ satisfy $RR^\top = I_s$. Define*

$$\mathcal{S}_R := \{UR : U \in \mathbb{R}^{d \times s}\}.$$

Then for every $U \in \mathbb{R}^{d \times s}$,

$$F_k(UR) = F_s(U), \quad (22)$$

and

$$\partial_C F_k(UR) = \{DR : D \in \partial_C F_s(U)\}. \quad (23)$$

Consequently, $\mathcal{S}_R$ is invariant under every Clarke-subgradient update, and the induced dynamics are exactly the rank- s dynamics.

Proof. Since $RR^\top = I_s$, $(UR)(UR)^\top = UU^\top$, proving (22). The residuals at UR and U are identical. For any common admissible sign vector and the corresponding matrix G from Theorem 2.2,

$$G(UR) = (GU)R.$$

This proves both inclusions in (23) and the update claim. $\square$

Corollary 5.2 (Conditional strict-saddle lifting). *Let $s = r$ and suppose an exact-rank trajectory satisfies*

$$U_t \longrightarrow U_\infty, \quad U_\infty U_\infty^\top = X_\star.$$

For every $R \in \mathbb{R}^{r \times k}$ with $RR^\top = I_r$, the lifted trajectory $W_t = U_t R$ converges to $W_\infty = U_\infty R \in \mathcal{W}_k$. If W_∞ is a strict saddle of F_k , this is a deterministic SubGM trajectory converging to a strict saddle.

5.1 A verified Gaussian instantiation from published theorems

The following statement is deliberately phrased at the level of the published hypotheses, so that no unproved compatibility assumption is hidden.

Theorem 5.3 (Gaussian exceptional trajectories). *Consider the common symmetric Gaussian sensing model used in [3, 1]: the A_i have independent standard Gaussian entries, the corruption support is independent, and the nonzero corruptions obey the noise assumptions of [1, Corollary 1]. Fix a corruption probability $p < 1/2$. Choose the ℓ_1/ℓ_2 -RIP tolerance $\delta > 0$ sufficiently small that*

$$p < \frac{1}{2} - \frac{\delta}{\sqrt{2/\pi} - \delta}, \quad (24)$$

and choose k/r and the sample constants so that the sample window

$$C_{\text{loc}}(\delta, p) dr \leq m \leq c_{\text{sad}}(p) dk \quad (25)$$

falls simultaneously within the exact-rank local-convergence theorem of [3] and the strict-saddle corollary of [1]. Then, with probability tending to one in the corresponding asymptotic regime, the same sensing instance has both properties:

- (i) *all rank- k true factors are Clarke-critical strict saddles;*
- (ii) *there is a nonempty local basin in the exact-rank problem and an admissible geometric SubGM schedule whose trajectories converge exactly to the rank- r true-factor set.*

Every instance in this joint event therefore admits an overparameterized rank- r trajectory converging to a strict-saddle true factor.

Proof. For the chosen δ and p , the Gaussian ℓ_1/ℓ_2 -RIP theorem of [3] applies to the full measurement operator and, under independent Bernoulli corruption, to the clean suboperator, with high probability once $m \geq C_{\text{loc}} dr$. Their sharpness/weak-convexity result and geometric-step SubGM theorem then yield exact local convergence for the rank- r factorization. Their corruption values may be arbitrary on the corrupted support, so the additional nondegeneracy assumptions of [1] are compatible with this result.

On the other hand, the symmetric Gaussian strict-saddle corollary of [1] applies with high probability in a window of the form $Cdr \leq m \leq cdk$ once k is a sufficiently large multiple of r (with constants depending on the fixed corruption parameters). Choose k/r large enough that (25) is nonempty. A union bound shows that the two high-probability events occur simultaneously. Finally apply Theorem 5.2. $\square$

Remark 5.4 (What this theorem does not say). *The local exact-rank initializer in Theorem 5.3 is near the rank- r solution set; it is not asserted to have small norm, and it lies in a measure-zero subset of the ambient rank- k space. The theorem is an existential compatibility result, not a positive-probability random initialization theorem.*

5.2 An explicit two-by-two strict-saddle trajectory

The next construction verifies every assertion directly and also repairs the phrase “small exact-rank initialization” in a deterministic, non-Gaussian setting.

Theorem 5.5 (Arbitrarily small exact-rank initialization converging to a strict saddle). *Let $d = k = 2$, $r = 1$, and*

$$X_\star = e_1 e_1^\top, \quad A_1 = A_2 = \text{diag}(1, 0), \quad A_3 = I_2,$$

with measurements

$$y_1 = y_2 = 1, \quad y_3 = 1 + \varepsilon, \quad \varepsilon > 0.$$

Thus the third measurement is corrupted, giving a one-third corruption rate. Let $W_\star = \text{diag}(1, 0)$. Then:

- (i) $W_\star$ is a Clarke-critical strict saddle of F_2 ;
- (ii) for every $0 < \eta < 1/2$ and $0 < \rho < 1$, define

$$\Pi_\infty(\eta, \rho) := \prod_{t=0}^{\infty} (1 + 2\eta\rho^t), \quad x_0 := \Pi_\infty(\eta, \rho)^{-1}, \quad (26)$$

and initialize $W_0 = \text{diag}(x_0, 0)$. Ordinary SubGM with $\alpha_t = \eta\rho^t$ and the unique subgradient along the trajectory satisfies $W_t \rightarrow W_\star$;

- (iii) for fixed $\eta > 0$, the norm $\|W_0\|_F = x_0$ can be made arbitrarily small by taking ρ sufficiently close to one.

Proof. At $W_\star$, the residuals are $0, 0, -\varepsilon$. Choose Clarke signs $s_1 = s_2 = 1/2$ and $s_3 = -1$. Since the A_i are symmetric,

$$G_\star = \frac{2}{3} \left(\frac{1}{2}A_1 + \frac{1}{2}A_2 - A_3 \right) = \text{diag}(0, -2/3),$$

so $G_\star W_\star = 0$. Hence $0 \in \partial_C F_2(W_\star)$. For the unit-Frobenius direction $V = \text{diag}(0, 1)$ and $0 < \gamma < \sqrt{\varepsilon}$,

$$F_2(W_\star + \gamma V) - F_2(W_\star) = \frac{1}{3}(|\gamma^2 - \varepsilon| - \varepsilon) = -\frac{\gamma^2}{3}.$$

Thus $W_\star$ is a strict saddle.

Now suppose $W_t = \text{diag}(x_t, 0)$ with $0 < x_t < 1$. All three residuals are strictly negative, so the signs are uniquely $s_1 = s_2 = s_3 = -1$. Therefore

$$G_t = -\frac{2}{3}(A_1 + A_2 + A_3) = \text{diag}(-2, -2/3)$$

and

$$x_{t+1} = (1 + 2\eta\rho^t)x_t. \quad (27)$$

The infinite product in (26) is finite and positive because $\sum_t \eta\rho^t < \infty$. Induction gives

$$x_t = \frac{\prod_{j=0}^{t-1} (1 + 2\eta\rho^j)}{\prod_{j=0}^{\infty} (1 + 2\eta\rho^j)}.$$

Hence $0 < x_t < 1$ for every finite t , validating the sign computation, and $x_t \uparrow 1$. Thus $W_t \rightarrow W_\star$.

Finally, given N , choose ρ so close to one that $\rho^{N-1} \geq 1/2$. The first N factors satisfy $1 + 2\eta\rho^t \geq 1 + \eta$, so

$$\Pi_\infty(\eta, \rho) \geq (1 + \eta)^N.$$

Letting $N \rightarrow \infty$ shows that x_0 can be made arbitrarily small. $\square$

Remark 5.6. *The construction is deliberately transparent. It proves that strict saddle geometry and deterministic convergence from an arbitrarily small rank- r initialization can coexist. It does not give a full-dimensional basin: every W_t has rank one, exactly as predicted by [Theorem 3.1](#).*

5.3 The exceptional set is null

Proposition 5.7 (Rank- r initializations form a null set). *If $r < \min\{d, k\}$, the determinantal variety*

$$\mathcal{R}_r := \{W \in \mathbb{R}^{d \times k} : \text{rank}(W) \leq r\}$$

has dimension $r(d + k - r) < dk$ and Lebesgue measure zero. In particular, each invariant face $\mathcal{S}_R$ and the union of all rank- r faces are null.

Proof. Every $(r + 1) \times (r + 1)$ minor vanishes on $\mathcal{R}_r$, and at least one such minor is a nonzero polynomial on $\mathbb{R}^{d \times k}$. Thus $\mathcal{R}_r$ lies in the zero set of a nonzero polynomial and has Lebesgue measure zero. The standard dimension formula is $r(d + k - r)$, and $dk - r(d + k - r) = (d - r)(k - r) > 0$. $\square$

6 A short asymmetric extension

The same obstruction is not peculiar to the symmetric parameterization. For $U \in \mathbb{R}^{d_1 \times k}$ and $V \in \mathbb{R}^{d_2 \times k}$, consider

$$F^{\text{asy}}(U, V) := \frac{1}{m} \sum_{i=1}^m \left| \left\langle A_i, UV^\top \right\rangle - y_i \right|. \quad (28)$$

For admissible signs define $Q_t = m^{-1} \sum_i s_{i,t} A_i$ and stack $Z_t = [U_t^\top, V_t^\top]^\top$.

Proposition 6.1 (Stacked multiplicative dynamics). *Every Clarke-subgradient update for (28) has the form*

$$Z_{t+1} = \left[I - \alpha_t \begin{pmatrix} 0 & Q_t \\ Q_t^\top & 0 \end{pmatrix} \right] Z_t. \quad (29)$$

The block matrix in (29) has operator norm $\|Q_t\|_{\text{op}}$. Hence the product theorem applies whenever the effective increments are summable and nonsingular. In particular, for the normalized rule $\alpha_t = \eta\rho^t / \|Q_t\|_{\text{op}}$, rank of the stacked factor is preserved if $0 < \eta < 1$.

Proof. The subgradients with respect to U and V are $Q_t V_t$ and $Q_t^\top U_t$, respectively, giving (29). The symmetric block matrix has eigenvalues equal to the signed singular values of Q_t , and therefore has operator norm $\|Q_t\|_{\text{op}}$. $\square$

Corollary 6.2 (Balanced true factors). *Suppose a true pair $(U_\star, V_\star)$ is balanced, $U_\star^\top U_\star = V_\star^\top V_\star$, and $\text{rank}(U_\star V_\star^\top) = r$. Then the stacked matrix $Z_\star = [U_\star^\top, V_\star^\top]^\top$ has rank r . Consequently, a stacked initialization of rank different from r cannot converge in factor space to such a balanced true pair under the hypotheses of [Theorem 3.1](#).*

Proof. Let $S = U_\star^\top U_\star = V_\star^\top V_\star$ and let $s = \text{rank}(S)$. On the support of S , polar decompositions give $U_\star = PS^{1/2}$ and $V_\star = RS^{1/2}$ with P and R isometric. Thus $U_\star V_\star^\top = PSR^\top$ has rank s , so $s = r$. Moreover $Z_\star^\top Z_\star = 2S$, hence $\text{rank}(Z_\star) = r$. Apply [Theorem 3.1](#). $\square$

Remark 6.3. *Unlike the symmetric case, an asymmetric product $UV^\top$ can have rank r even when the stacked factor has larger rank, through unbalanced cancellations. Therefore [Theorem 6.2](#) rules out convergence to the balanced true factors used in standard SVD parameterizations; it does not rule out every possible high-factor-rank representation of the same matrix.*

7 What is solved and what remains open

7.1 Rigorous progress obtained here

For symmetric robust matrix sensing, the paper proves the following.

- (1) **Exact asymptotic obstruction.** Under summable effective increments and nonsingular multipliers, a full-rank initialization cannot converge exactly to the rank- r true-factor set. This includes ordinary geometric SubGM with $\eta L_A < 1$ and the normalized exponential rule of [\[2\]](#) with $\eta < 1/2$.
- (2) **Quantitative obstruction.** The trailing singular values of the initialization give explicit, uniform positive lower bounds on both factor and matrix error.
- (3) **Exceptional positive examples.** Rank- r invariant faces can carry deterministic trajectories to strict saddles. This is verified in an overlapping Gaussian theorem regime and in the explicit 2×2 construction of [Theorem 5.5](#).
- (4) **Measure-theoretic classification.** Every such rank- r construction lies inside a determinantal variety of Lebesgue measure zero.

These conclusions repair the mathematically valid part of the proposed resolution and sharpen it. They also explain why the reported finite precision trajectories may appear to converge even though exact full-rank convergence is impossible under the stated small-step conditions.

7.2 Questions not resolved

The following remain open after this analysis.

- (1) Whether the true factors are active or non-active strict saddles in the sense of [\[4\]](#). The multiplicative/rank argument does not construct or exclude an active manifold.
- (2) Positive-measure attraction to an ε -neighborhood, finite-time statistical recovery, and the behavior of floating-point implementations.
- (3) Nonsummable steps, stochastic perturbations, proximal algorithms, or trajectories containing a singular update matrix. Any of these can evade the product theorem.
- (4) In the asymmetric problem, exact matrix convergence through unbalanced high-rank factor representations.

Thus the appropriate classification is

partial progress, not a complete solution of the authors' question.

8 Audit of the original proposed resolution

Proposed claim	Defect	Correct statement
Positive probability over sensing instances contradicts almost-sure saddle avoidance.	Saddle avoidance concerns measure over initialization, usually conditional on an instance.	The exact-rank basin lies in a null determinantal variety; under the small summable rules considered here, full-dimensional exact convergence has probability zero.
The face $[U, 0]$ is invariant and hence attracting.	Invariance is not transverse attraction.	For $W_t = [U_t, Z_t]$, both blocks are multiplied by the same invertible product; in fact $\ P_t Z_0\ _F \geq c_H \ Z_0\ _F$.
A standard small full-rank initialization converges exactly.	The cited small-initialization result tunes the scale to a requested finite accuracy.	A fixed higher-rank initialization has the explicit floors (19)–(20).
A deterministic strict-saddle trajectory proves a non-active saddle.	A measure-zero stable trajectory is compatible with active as well as non-active geometry.	Exceptional rank- r convergence is real, but the active/non-active classification remains unresolved.

Table 1: Logical corrections to the original argument.

9 Conclusion

The key identity $D(W) = G(W)W$ has two complementary implications. It creates invariant rank- r faces, allowing specially initialized deterministic trajectories to converge to overparameterized strict saddles. At the same time, for summable nonsingular updates it prevents a full-rank trajectory from shedding its redundant directions, even at the limit. The latter fact supplies a quantitative exact-recovery obstruction that applies directly to the exponentially decaying rules used in the motivating work under explicit small-step conditions.

Accordingly, the original invariant-face manuscript is not a solution of the authors’ open problem. Its valid core can be repaired into a rigorous exceptional-trajectory theorem, and the stronger rank-preservation theorem provides substantive additional progress. The active/non-active strict-saddle question and positive-measure approximate attraction remain open.

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